Digital Twin for Plant Health Monitoring - update

Rayan Contuliano Bravo - February 2026

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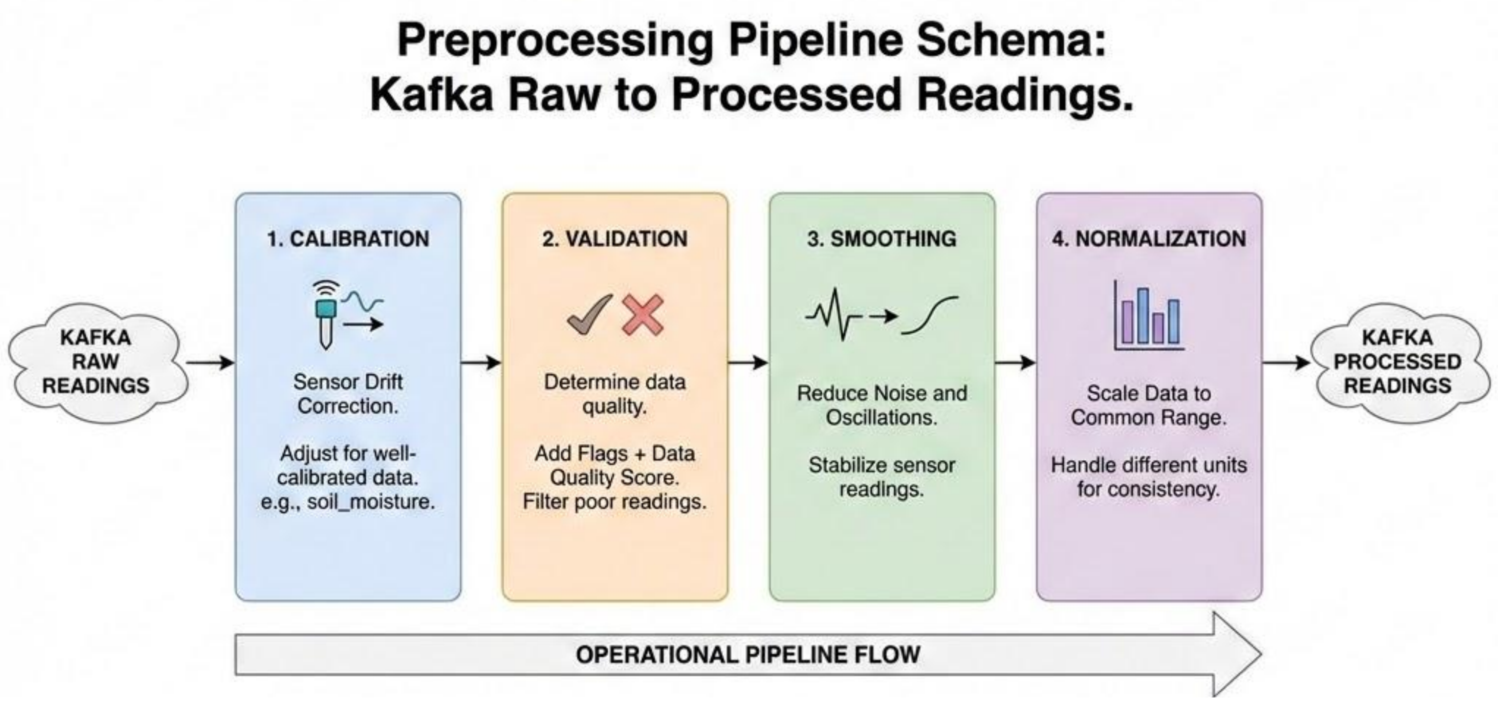
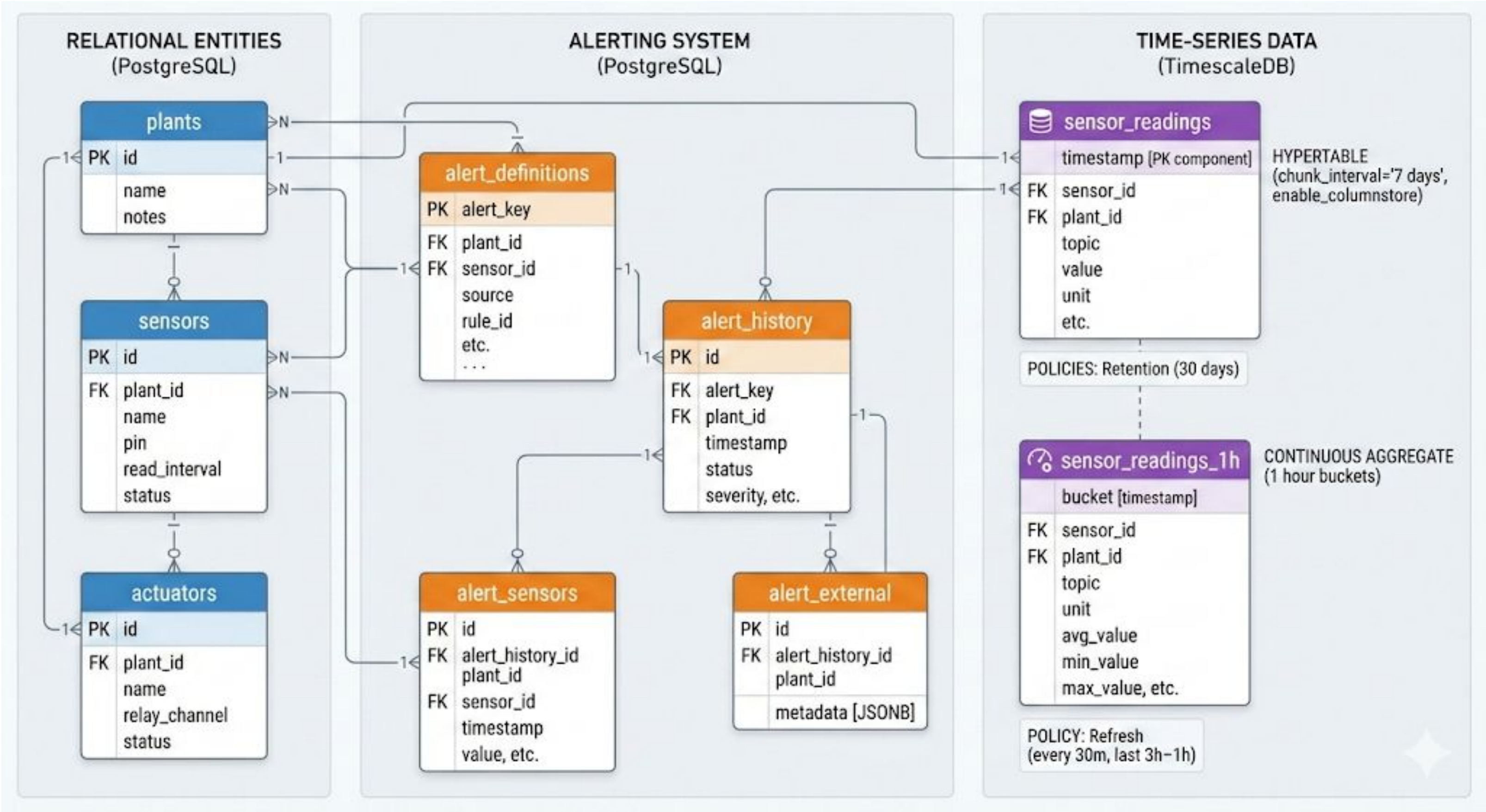
Preprocessing module (Spark Structured Streaming)



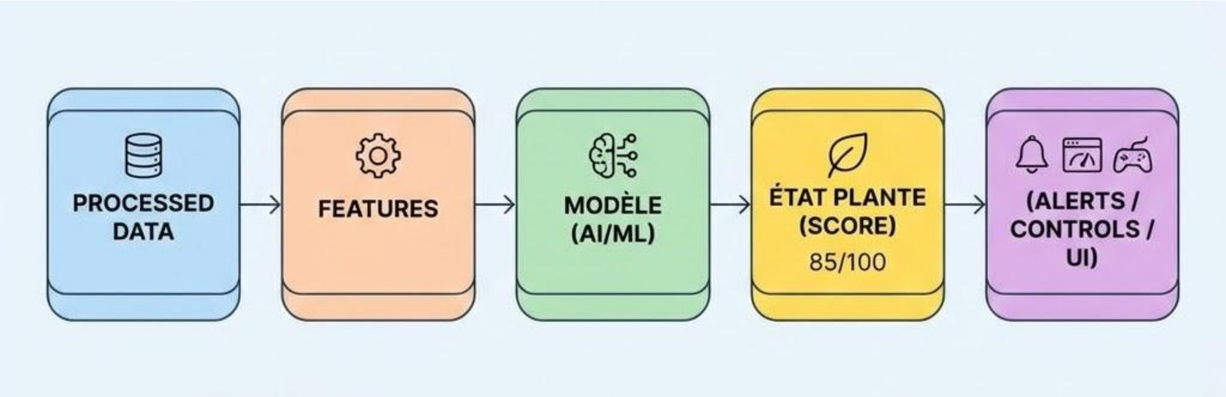
Database module (PostgreSQL & TimescaleDB)

- Replaced InlfuxDB by PostgreSQL with the TimescaleDB extension
  - Better for handling both time-series data and relational data
  - SQL syntax
  - Down sampling and retention policies
- Other modules can query the Database by using the REST API of the module


RELATIONAL ENTITIES
(PostgreSQL)
ALERTING SYSTEM
(PostgreSQL)
TIME-SERIES DATA
(TimescaleDB)
plants
PK id
name
notes
sensors
PK id
FK plant_id
name
pin
read_interval
status
actuators
PK id
FK plant_id
name
relay_channel
status
alert_definitions
PK alert_key
FK plant_id
FK sensor_id
source
rule_id
etc.
alert_history
PK id
FK alert_key
FK plant_id
timestamp
status
severity, etc.
alert_sensors
PK id
FK alert_history_id
plant_id
FK sensor_id
timestamp
value, etc.
alert_external
PK id
FK alert_history_id
plant_id
metadata [JSONB]
sensor_readings
timestamp [PK component]
FK sensor_id
FK plant_id
topic
value
unit
etc.
HYPERTABLE
(chunk_interval='7 days',
enable_columnstore)
POLICIES: Retention (30 days)
sensor_readings_1h
bucket [timestamp]
FK sensor_id
FK plant_id
topic
unit
avg_value
min_value
max_value, etc.
CONTINUOUS AGGREGATE
(1 hour buckets)
POLICY: Refresh
(every 30m, last 3h–1h)

Alerting System

- Based on processed sensor readings
- User can set alerting preference (thresholds, cooldown to avoid spams, …)
- Would like to use this base alerting module to build the analytics module
    - State of the plant
    - Features
    - …



Controller module

- Actuators managed by a relay
- Allow the user to create routines via the web application
  - Routines are executed by a scheduler
  - Can be triggered by a sensor reading or based on time
- Routines not yet operational but actuators are actionable

Web application

- Now using Svelte, Tailwind and typescript :
    - Svelte: Easier to design components and better reactivity
    - Tailwind: Easier to stylize
    - Typescript: Javascript but with typing so less runtimes bug and more compile bugs :-)
- Little bit more complex but easier to maintain

Next steps

- Finish controller module by handling the setup of routines
- Handling RPI Camera
- Implementing the Analytics module based on the alerting system

DIGITAL TWIN

CONNECTED

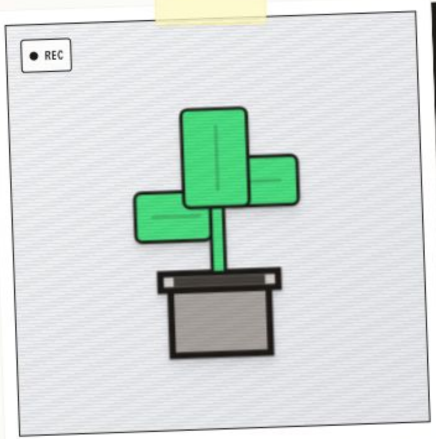
OVERVIEW | ANALYTICS | JOURNAL

Last update: 10:10 AM

Basil Study

Current Status: Cold & Dying

PIXEL | PHOTO



My workspace Buddy

10:10 AM

FAN	HEATER	LIGHT	PUMP
OFF	OFF	OFF	OFF

VITALITY

98% Excellent

TEMPERATURE

16.6°C

ROOM AMBIENT

Cold

HUMIDITY

63%

AIR SENSOR

Humid

MOISTURE

71%

SOIL SENSOR A

Wet Soil

LIGHT

54lx

WINDOW SENSOR

Low Light

ROUTINES

AI AUTO-PILOT

+ NEW RULE

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Last update: 10:05 AM

Sensor Trends

Historical data for Basil Study

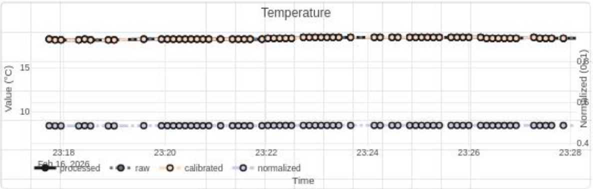
DAY WEEK MONTH

PROCESSED RAW CALIBRATED NORMALIZED

DQ (hover): –

Temperature

AVG: 23.5°C

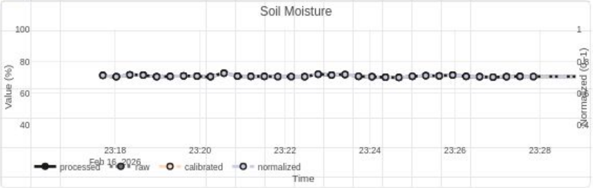


Humidity

AVG: 46%

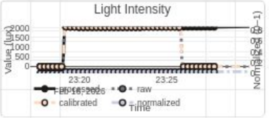
Soil Moisture

AVG: 45%



Light Exposure

TOTAL: 8H 12M



Data Freshness

LIVE

Heartbeat

Updated 2s ago

Connection stability: 99.9%

Packets received: 14,203

AI Insights

- Temperature peaks at 2PM usually correspond with direct sunlight.
- Soil moisture depletion rate is stable. Next water cycle in approx 2 days.

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Last update: 10:05 AM

Alerts & Journal

Attention Needed: 4 Active Alerts

EXPORT LOGS

Active Crises

Data quality score below 0.7
info • active
DISMISS

Data quality score below 0.7
info • active
DISMISS

Light level below 1000.0 lux
warning • active
DISMISS

Data quality score below 0.7
info • active
DISMISS

Water Tank Status

REFILL

Level 3.5L

REFILLED
2d Ago

EMPTY IN
4 Days

MANUAL LOGS

Today
10:42 AM

Manual Mist

Humidity was reading low on sensor A, gave a quick spritz to the upper leaves.

Yesterday
04:15 PM

Pruning Session

Removed two yellowing leaves near the base. Plant looks much tidier. Checked for pests - all clear.

#maintenance #health-check

Oct 24
09:00 AM

Fertilizer Added

Added 5ml of liquid fertilizer to the water tank. Regular monthly feeding schedule.

ENTRY TITLE: Entry title
TAGS: Tags (comma separ
ICON: Note
COLOR: Gray

Write a new journal entry...

SAVE